



Evaluation Table and final results

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Below is a **very solid evaluation table structure** commonly used in papers comparing **rule learners, subgroup discovery algorithms, and tree models**. It separates **predictive performance, rule structure, and interpretability**, which is exactly what your thesis needs.

1 Model Performance Comparison

This table evaluates **predictive ability**.

Model	Target type	RMSE	MAE	R ²	Accuracy (if classification)
CarenR	Continuous	0.82	0.63	0.71	—
RIPPER	Discretized	—	—	—	0.78
Regression Tree	Continuous	0.79	0.61	0.73	—
Random Forest (optional)	Continuous	0.68	0.52	0.81	—

Interpretation

- Tree / RF → predictive upper bound
- CarenR → interpretable regression-style rules
- RIPPER → interpretable classification rules

2 Rule Structure Comparison

This is **very important for your thesis** because you are comparing rule discovery.

Model	# Rules	Avg Rule Length	Max Rule Length	Coverage (%)	Overlap
CarenR	6	3.1	5	74	low
RIPPER	8	3.7	6	70	moderate
Tree (paths)	12	4.9	7	100	high

Definitions

Number of rules

Total rules produced.

Average rule length

Average number of conditions per rule.

Coverage

Percentage of dataset explained by rules.

Overlap

How much rules cover the same observations.

Rule learners usually aim for:

- **few rules**
- **short rules**
- **high coverage**

3 Rule Quality Metrics

These measure **rule strength**.

Model	Avg Support	Avg Confidence	Avg Lift	Avg WRAcc
CarenR	0.18	0.74	1.36	0.11
RIPPER	0.16	0.71	1.31	0.09
Tree rules	0.09	0.79	1.28	0.06

Metrics

Support

Proportion of dataset covered.

Confidence

Accuracy of rule predictions.

Lift

Improvement vs baseline.

WRAcc (Weighted Relative Accuracy)

Very common metric in **subgroup discovery**.

4 Structural Pattern Comparison

This table checks if algorithms find **similar patterns**.

Pattern	CarenR	RIPPER	Tree
High temp + low rainfall → high production	✓	✓	✓
Early flowering + dry summer → low production	✓	✓	partial

Pattern	CarenR	RIPPER	Tree
High humidity → reduced yield	✓	✗	✓

This is where you show:

- convergence of patterns
- differences between methods

This section is often the **most interesting part of a rule discovery thesis**.

5 Model Complexity Comparison

Model	Parameters	Complexity	Interpretability
CarenR	6 rules	Low	Very High
RIPPER	8 rules	Low	High
Regression Tree	12 leaves	Medium	Medium
Random Forest	500 trees	High	Low

6 Suggested Final Evaluation Table

This is a **summary table** many theses include.

Model	Prediction Quality	Interpretability	Rule Simplicity	Target Type
CarenR	Medium–High	Very High	High	Continuous
RIPPER	Medium	High	Medium	Discretized
Regression Tree	High	Medium	Medium	Continuous
Random Forest	Very High	Low	Low	Continuous

7 Why this structure is strong for your thesis

It separates evaluation into **three dimensions**:

Prediction

Are the models accurate?

Structure

Are the rules simple and meaningful?

Discovery

Do different algorithms identify similar phenomena?

This aligns perfectly with your thesis goal:

- discovering interpretable rules for wine production

- comparing rule discovery methods
 - validating findings with predictive models
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8 One extra table I strongly recommend

Because your dataset spans **1933–2022**, rule stability is important.

Model	Rule Stability	Comment
CarenR	High	Rules consistent across folds
RIPPER	Medium	Some variability
Tree	Low	Structure changes significantly

This makes your analysis much stronger.

If you want, I can also show you **one extremely powerful analysis most theses miss**:

Rule overlap analysis between CarenR, RIPPER, and trees, which visually proves whether the algorithms discovered the **same climate-production mechanisms**.